

Software Modeling and Design Project

Topic: Food Ordering and Delivering System

Group: A

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User Levels

Primary Users

1. Customer

The customer is the main user who orders food through the application. Customers use the system to browse restaurants, select food items, place orders, and track their deliveries.

Responsibilities:

- Register and log into the system
- Browse restaurants and food menus
- Search for restaurants or food items
- Add food items to the cart
- Place food orders
- Make payments for orders
- Track order and delivery status
- View order history
- Rate and review restaurants and food

2. Restaurant Staff

Restaurant staff manage food orders received from customers and prepare the food for delivery.

Responsibilities:

- Log into the system
- Manage restaurant profile information
- Add, update, or remove menu items
- Receive customer orders
- Accept or reject orders
- Prepare food orders
- Update order status (accepted, preparing, ready)
- Manage food availability and pricing

3. Delivery Driver

The delivery driver is responsible for delivering food orders from restaurants to customers.

Responsibilities:

- Log into the system
- View assigned delivery requests
- Accept delivery tasks
- Pick up food orders from restaurants
- Deliver orders to customers
- View delivery addresses and order details
- Update delivery status (picked up, on the way, delivered)
- Confirm order delivery

Secondary Users

4. Administrator (Admin)

The administrator manages the platform and ensures that the system operates correctly.

Responsibilities:

- Manage customer accounts
- Manage restaurant accounts
- Manage delivery driver accounts
- Monitor system activity
- Manage orders and platform data
- Handle user complaints and reports
- Generate system reports
- Ensure system security and reliability

5. Finance Manager

The finance manager manages all financial operations related to the platform.

Responsibilities:

- Monitor customer payments
- Manage financial transactions
- Track platform revenue
- Generate financial reports
- Verify payment records
- Handle payment-related issues

6. Customer Support

Customer support assists users when they encounter problems while using the system.

Responsibilities:

- Respond to customer inquiries
- Assist users with order problems
- Resolve payment issues
- Handle complaints and feedback
- Communicate with restaurants and delivery drivers
- Provide help for using the application

7. Marketing Manager

The marketing manager is responsible for managing promotional activities and campaigns on the platform.

Responsibilities:

- Manage marketing campaigns and promotional activities
- Manage promotions, discounts, and coupon codes
- Handle collaboration requests from restaurants
- Send promotional notifications to users
- Monitor and analyze campaign performance
- Manage advertisements and featured restaurants

External Users

8. System Administrator / IT Support

The system administrator manages the technical infrastructure and ensures the system runs smoothly.

Responsibilities:

- Maintain system servers and databases
- Monitor system performance
- Ensure system security
- Perform system updates and maintenance
- Fix technical issues and system errors
- Manage backups and data recovery

9. Payment System

The payment system is an external service that processes online payments made by customers. It ensures secure financial transactions between the customer and the platform.

Responsibilities:

- Process online payments for orders
- Verify payment information securely
- Confirm successful or failed transactions
- Handle payment authorization and validation
- Communicate payment results to the platform

10. Notification System

The notification system is responsible for sending system messages and updates to users. It helps inform customers, restaurants, and delivery drivers about important events in the system.

Responsibilities:

- Send order confirmation notifications
- Send order status updates
- Send delivery status notifications
- Send system alerts and reminders
- Notify users about successful or failed payments

2. Functional Requirements

Customer

1. Customer manages account (register, login, update profile)
2. Customer manages authentication and security settings
3. Customer browses, searches, filters, and sorts restaurants
4. Customer views restaurant and food item details (menu, ratings, information)
5. Customer manages shopping cart (add, remove, update items)
6. Customer places orders (reviews cart, selects delivery address, payment method, and completes payment)
7. Customer tracks order status in real time and receives notifications
8. Customer views order history and can reorder previous orders
9. Customer rates and reviews restaurants
10. Customer manages personalized experience (preferences, favorites, recommendations)
11. Customer interacts with the platform for support, promotions, and communication
12. Customer manages payment history and transaction records
13. Customer manages refunds, complaints, and issue requests

Restaurant Staff

14. Restaurant staff manages account and restaurant profile information (login, profile management)
 15. Restaurant staff manages menu items, categories, and food availability (add, update, remove, organize)
 16. Restaurant staff manages customer orders and updates order status (receive, view, accept, reject, preparing, ready for delivery)
 17. Restaurant staff monitors and analyzes restaurant performance
 18. Restaurant staff manages temporary service interruptions (e.g., busy, closed)
 19. Restaurant staff views sales analytics (daily, weekly, monthly)
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Delivery Driver

20. Delivery driver manages account and profile (login, profile management)
 21. Delivery driver manages delivery requests and details (view, accept, reject, view details)
 22. Delivery driver completes deliveries (navigate, pick up, deliver orders, update and confirm delivery status)
 23. Delivery driver monitors personal performance and delivery outcomes
 24. Delivery driver manages delivery issues and exceptions (e.g., incorrect address, unavailable customer)
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Administrator

25. Administrator manages account and profile (login, profile management)
 26. Administrator manages user accounts (customers, restaurants, drivers)
 27. Administrator monitors system activity
 28. Administrator manages orders in the system
 29. Administrator handles complaints and support requests
 30. Administrator manages platform settings
 31. Administrator generates system reports
 32. Administrator manages account approvals and restrictions (approve/reject restaurant registrations, suspend/ban user accounts)
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Finance Manager

33. Finance manager manages account and profile (login, profile management)
34. Finance manager monitors transactions and manages billing, payouts, refunds, and chargebacks
35. Finance manager generates and analyzes financial reports and forecasting data

36. Finance manager manages financial operations (commissions, taxes, promotions, and external integrations)
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Customer Support

37. Customer support manages account and profile (login, profile management)
38. Customer support manages user support operations (handles inquiries, complaints, issue resolution, and communication with users and service providers)
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Marketing Manager

39. Marketing manager manages account (login, profile management)
40. Marketing manager manages marketing collaboration requests (receives, reviews, approves, or rejects requests from restaurants)
41. Marketing manager manages marketing campaigns (creates, schedules, targets customer segments, and manages promotions for approved restaurants)
42. Marketing manager manages promotions, discounts, and coupon codes (for selected restaurants or platform-wide)
43. Marketing manager manages promotional communication (sends notifications and promotional messages to users)
44. Marketing manager analyzes and monitors campaign performance and customer engagement metrics
45. Marketing manager manages featured restaurants and advertisements for approved partners

3. Non-Functional Requirements

Performance

1. Performance Requirement

The system shall process user requests and display results within 2 seconds under normal operating conditions.

2. Response Time Requirement

Restaurant menus and food items shall be displayed within 3 seconds after a user request.

3. Concurrent Users Requirement

The system shall support at least 500 concurrent users placing orders simultaneously.

4. System Capacity Requirement

The system shall be capable of handling thousands of orders per day without performance degradation.

5. Notification Performance Requirement

The system shall send order confirmation and delivery notifications within a few seconds after order status changes.

6. Order Tracking Requirement

The system shall update order and delivery status in real time or near real time.

Security Requirements

7. Security Requirement

The system shall protect user accounts and payment information through secure authentication mechanisms and encrypted data storage.

8. Authentication Requirement

The system shall require users to authenticate (register and log in) before accessing personalized services.

9. Authorization Requirement

The system shall restrict access to system functionalities based on user roles such as Customer, Restaurant Staff, Delivery Driver, and Administrator.

10. Privacy Requirement

The system shall ensure that user personal data is confidential and protected from unauthorized access.

11. Payment Security Requirement

The system shall ensure secure communication with external payment systems during financial transactions.

Reliability and Data Management

12. Reliability Requirement

The system shall ensure that all orders are processed correctly without duplication or loss.

13. Data Integrity Requirement

The system shall maintain accurate and consistent data for users, orders, and payment transactions.

14. Backup Requirement

The system shall perform automatic database backups on a daily basis to prevent data loss.

15. Recovery Requirement

The system shall be able to restore system data from backups in case of system failure.

16. Data Storage Requirement

The system shall securely store user profiles, order history, and transaction records.

Usability and Accessibility

17. Usability Requirement

The system interface shall be simple, intuitive, and easy to navigate for all types of users.

18. Accessibility Requirement

The system shall be accessible through web browsers and mobile devices with an internet connection.

19. Error Handling Requirement

The system shall display clear and informative error messages when problems occur.

Compatibility and Scalability

20. Compatibility Requirement

The system shall function correctly on major web browsers such as Chrome, Firefox, and Edge.

21. Scalability Requirement

The system shall support growth in users, restaurants, and delivery drivers without performance degradation.

Maintainability and Monitoring

22. Maintainability Requirement

The system shall be developed using a modular architecture to simplify system maintenance and updates.

23. Logging Requirement

The system shall maintain activity logs of user actions, orders, and system operations.

24. System Monitoring Requirement

The administrator shall be able to monitor system activity and platform performance.

Availability

25. Availability Requirement

The system shall be available 24 hours a day, 7 days a week, with a minimum uptime of 99%.